Supplementary Material for: Frontier and Localhost: A Survey of How Production AI Learns Outside the Weights

October 2026

This document accompanies the main article. S1 is the released score table; S2–S5 carry the verification material of the survey (data appendix, mechanism stack, contrast set and matrix membership, and the per-system walkthrough of the convergent research families); S6 is the audit of every absolute claim of the earlier preprint against the October-2026 literature, with the rubric-reward comparison that Section IV of the main text summarizes. Section numbers in the form "Section N" refer to the main article.

S1. Full score table

Every scored (substrate, system) row of the corpus: 130 systems scored by May 2026 and 126 added by the October 2026 sweep. Scores follow the 0–5 rubric of the main text (Table V-1); Loop 1 / Loop 2 gate types and signal classes are recorded for the October rows (and for May rows re-audited in Section VII). Substrate codes: INT integrated, S1 instructions, S2 skills, S3 memory, S4 tools, S5 orchestration, S6 governance. The CSV files with per-row justifications accompany this document.

Sub.	System	Score	Year	Source	Loop 1	Loop 2	Class	Added
INT	ADAS / Meta Agent Search	4	2024	arXiv:2408.08435				May 26
INT	Agent S / S2	4.5	2024	arXiv:2410.08164				May 26
INT	Agent Workflow Memory (AWM)	3.5	2024	arXiv:2409.07429				May 26
INT	AlphaEvolve	4	2025	arXiv:2506.13131				May 26
INT	AutoAgent	5	2026	arXiv:2603.09716				May 26
INT	AutoManual	3.5	2024	NeurIPS 2024				May 26
INT	AutoSaddler	4	2026	arXiv:2608.23041	Auto	None	C2	Oct 26
INT	CaptainAgent	2.5	2024	arXiv:2405.19425				May 26
INT	CASCADE	4.5	2025	arXiv:2512.23880				May 26
INT	ChatDev	3	2023	arXiv:2307.07924				May 26
INT	Co-Harness	4	2026	arXiv:2607.22688	Auto	None	C2	Oct 26
INT	CoEvoSkills	3.5	2026	arXiv:2604.01687				May 26
INT	Cognition Devin	3	2024	cognition.ai				May 26
INT	Continual Harness	4	2026	arXiv:2605.09998	None	None	C4	Oct 26
INT	Cradle	4	2024	arXiv:2403.03186				May 26
INT	Darwin Gödel Machine	4	2025	arXiv:2505.22954				May 26
INT	Double Ratchet	4	2026	arXiv:2607.12790	Auto	None	C3	Oct 26
INT	EvolveNet	4	2026	arXiv:2608.04968	Auto	Auto	an C1,C2	Oct 26
INT	EvolveR	4	2025	arXiv:2510.16079				May 26
INT	ExpeL	3.5	2023	arXiv:2308.10144				May 26
INT	Generative Agents	4	2023	arXiv:2304.03442				May 26
INT	GenericAgent	4	2026	arXiv:2604.17091	Auto	None	C2	Oct 26
INT	GITM	3	2023	arXiv:2305.17144				May 26
INT	HarnessBank	4	2026	arXiv:2607.13683	Auto	None	C2	Oct 26
INT	HarnessX	4	2026	arXiv:2606.14249	Auto	None	C2	Oct 26
INT	Hippocratic AI Polaris	3.5	2024	hippocraticai.com				May 26
INT	JARVIS-1	3.5	2023	arXiv:2311.05997				May 26
INT	JIT-Agent	4	2026	arXiv:2608.25593	Auto	None	C2	Oct 26
INT	Life-Harness	4	2026	arXiv:2605.22166	Auto	None	C2	Oct 26
INT	Mendel Gödel Machine (MGM)	4	2026	arXiv:2608.07645	Auto	None	C1	Oct 26
INT	MetaGPT	3.5	2023	arXiv:2308.00352				May 26
INT	Mobile-Agent-E	4.5	2025	arXiv:2501.11733				May 26
INT	ModularRSI	4	2026	arXiv:2609.14857	Auto	None	C2	Oct 26
INT	NanoResearch	5	2026	arXiv:2605.10813				May 26
INT	OpenCore	4	2025	github.com/sibmike/opencore				May 26
INT	OpenHands / CodeAct	3	2024	arXiv:2407.16741				May 26

Sub.	System	Score	Year	Source	Loop 1	Loop 2	Class	Added
INT	OS-Copilot / FRIDAY	4	2024	arXiv:2402.07456				May 26
INT	Plan4MC	2.5	2023	GitHub/arXiv				May 26
INT	SAGE	4	2025	arXiv:2512.17102				May 26
INT	Sakana AI Scientist	3	2024	arXiv:2408.06292				May 26
INT	ScholarEvolve	4	2026	arXiv:2609.40169	Auto	None	C2	Oct 26
INT	SIA	4	2026	arXiv:2605.27276	Auto	None	C2	Oct 26
INT	Sierra Agent OS	4	2024	sierra.ai				May 26
INT	Sierra Agent OS 2.0	3	2025	sierra.ai/blog/agent-os-2-0				May 26
INT	SkillForge	4	2026	arXiv:2604.08618				May 26
INT	SkillMAS	4	2026	arXiv:2605.09341	Auto	None	C2	Oct 26
INT	SkillOS	3	2026	SemanticScholar:2653e4c7				May 26
INT	SkillRL	5	2026	arXiv:2602.08234				May 26
INT	TTHE	4	2026	arXiv:2607.08124	Auto	None	C3	Oct 26
INT	VACE	4	2026	arXiv:2609.37105	Auto	None	C2	Oct 26
INT	Voyager	4	2023	arXiv:2305.16291				May 26
INT	Workspace Optimization (DreamTeam)	3	2026	arXiv:2605.09650	Auto	None	C1	Oct 26
S1	AGENTS.md	2	2026	arXiv:2601.20404				May 26
S1	Aider Conventions	2	2024	aider.chat				May 26
S1	Claude Code CLAUDE.md	4	2025	docs.anthropic.com				May 26
S1	Cody / Sourcegraph	2	2024	sourcegraph.com				May 26
S1	Continue.dev Rules	3	2024	docs.continue.dev				May 26
S1	Cursor Rules	3	2024	docs.cursor.com/rules				May 26
S1	GitHub Copilot Custom Instructions	3	2024	docs.github.com				May 26
S1	GRACE	4	2026	arXiv:2607.09175	Auto	None	C2	Oct 26
S1	Lovable Knowledge	2	2024	docs.lovable.dev				May 26
S1	Meta Context Engineering (MCE)	4	2026	arXiv:2601.21557	Auto	None	C2	Oct 26
S1	Replit replit.md	4	2025	docs.replit.com				May 26
S1	Spotify conversational-recommendation loop	4	2026	arXiv:2609.30297	Auto	Human	C3	Oct 26
S1	VS Code Copilot Instructions	3	2024	code.visualstudio.com				May 26
S1	Windsurf Cascade Rules	3	2024	docs.windsurf.com				May 26
S1	Zed AI Rules	3	2024	zed.dev/docs				May 26
S2	AgentVerse	1	2023	arXiv:2308.10848				May 26
S2	Anthropic Claude Agent Skills	3	2025	anthropic.com/engineering				May 26
S2	Anthropic/skills GitHub repo	2	2025	github.com/anthropics/skills				May 26
S2	ASSAY	4	2026	arXiv:2606.15390	Auto	None	C2	Oct 26
S2	AutoSkill	3	2026	arXiv:2603.01145	Auto	None	C3	Oct 26
S2	Buildrix	3	2026	arXiv:2606.25139	Human	Human	C3	Oct 26
S2	ChatGPT Plugins (legacy)	1	2023	openai.com				May 26
S2	Claude Code Skills	3	2025	docs.anthropic.com				May 26
S2	COSPLAY	5	2026	arXiv:2604.20987				May 26
S2	CrewAI Skills	2	2025	docs.crewai.com				May 26
S2	HiSME	4	2026	arXiv:2605.28390	Auto	None	C1,C3	Oct 26
S2	HuggingFace Agent Skills	2	2025	huggingface.co				May 26
S2	LangChain Skills	2	2025	langchain.com				May 26
S2	Library Drift curator	4	2026	arXiv:2605.19576	Auto	None	C1	Oct 26
S2	Meta Context Engineering (MCE)	4	2026	arXiv:2601.21557	Auto	None	C2	Oct 26
S2	MetaGPT (SOP-as-skill)	3	2023	arXiv:2308.00352				May 26
S2	MetaSkill-Evolve	4	2026	arXiv:2607.05297	Auto	None	C2	Oct 26
S2	Microsoft Semantic Kernel Plugins	2	2024	learn.microsoft.com				May 26
S2	MUSE-Autoskill	4	2026	arXiv:2605.27366	Auto	None	C1,C2	Oct 26
S2	NanoResearch Skill Bank	4	2026	arXiv:2605.10813				May 26
S2	OpenAI Custom GPTs / Actions	2	2023	platform.openai.com				May 26
S2	Personalized developer skills (Huang-Du-Lan)	3	2026	arXiv:2608.10319	Auto	None	C3	Oct 26
S2	RASO	4	2026	arXiv:2609.38024	Auto	Auto	C2	Oct 26
S2	SAGE	5	2025	arXiv:2512.17102				May 26
S2	SCAFFOLD	4	2026	arXiv:2609.05511	Auto	None	C2	Oct 26
S2	SE-GoS	4	2026	arXiv:2609.08228	Auto	None	C2	Oct 26
S2	SkeMex	4	2026	arXiv:2606.09365	Auto	None	C2	Oct 26
S2	Skill Issues (Bauplan)	4	2026	arXiv:2606.01185	Auto	None	C1,C3	Oct 26
S2	SKILL.md smells (Hong-Imani-Ahmed)	1	2026	arXiv:2607.01456	None	None	-	Oct 26
S2	SkillAudit	4	2026	arXiv:2606.14239	Auto	None	C3,C1	Oct 26
S2	SkillBoost	4	2026	arXiv:2607.26643	Auto	None	C1	Oct 26
S2	SkillChain	4	2026	arXiv:2606.12984	Auto->Huma	None	C3,C2	Oct 26
S2	SkillComposer	4	2026	arXiv:2606.06079	Auto	None	C1,C2	Oct 26
S2	SkillEvoReg	4	2026	arXiv:2609.30861	Auto	None	C1,C2	Oct 26
S2	SkillFoundry	3	2026	arXiv:2604.03964	Auto	None	C1	Oct 26
S2	SkillGen	4	2026	arXiv:2605.10999	Auto	None	C2	Oct 26
S2	SkillGLoW	4	2026	arXiv:2609.02217	Auto	None	C2	Oct 26
S2	SkillLift	4	2026	arXiv:2609.15396	Auto	None	C3 in	Oct 26
S2	SkillOps	4	2026	arXiv:2605.13716	Auto	None	C2	Oct 26
S2	SkillSmith	4	2026	arXiv:2606.01314	Auto	None	C1,C2	Oct 26
S2	SkillSpec (consensus-gated)	4	2026	arXiv:2610.00704	Auto	None	C1,C2	Oct 26
S2	SuperAGI Toolkits	1	2024	github.com/TransformerOptimu				May 26

Sub.	System	Score	Year	Source	Loop 1	Loop 2	Class	Added
S2	TRUSS	3	2026	arXiv:2608.17588	Auto	None	C1,C2	Oct 26
S2	VALVE	4	2026	arXiv:2609.32990	Auto	None	C1	Oct 26
S2	VCE-Skill	4	2026	arXiv:2608.16544	Auto	None	C1,C2	Oct 26
S2	Voyager	4	2023	arXiv:2305.16291				May 26
S2	Web2BigTable	3	2026	arXiv:2604.27221	Auto	None	C1,C2	Oct 26
S2	Who Maintains Agent Skills? (empirical)	2	2026	arXiv:2609.05677	Human	None	C3	Oct 26
S2	WikiSkill	4	2026	arXiv:2608.27454	Auto	None	C2	Oct 26
S2	XRepoSkill	4	2026	arXiv:2609.36807	Auto	Auto-parti	C1	Oct 26
S3	A-MEM	3	2025	NeurIPS 2025				May 26
S3	Agent-native memory system evaluation (Zhou et a	2	2026	arXiv:2606.24775	n/a	n/a	n/a	Oct 26
S3	AIM privacy-aware interoperable memory (Johnson	3	2026	arXiv:2609.12320	Auto	None	C3	Oct 26
S3	Beyond the Harness (enterprise Text-to-SQL cards	3	2026	arXiv:2608.22830	Auto	None	C2	Oct 26
S3	ChatGPT Memory	3	2024	openai.com				May 26
S3	Claude Memory	3	2025	anthropic.com				May 26
S3	Cognee	3	2024	github.com/topoteretes/cogne				May 26
S3	Compiled Memory (Rhodes & Kang)	4	2026	arXiv:2603.15666	Auto-gated	None	C1	Oct 26
S3	Control-plane placement / ForgetEval (Yang)	2	2026	arXiv:2606.15903	n/a	n/a	n/a	Oct 26
S3	FRESH failure-aware heterogeneous graph memory (	4	2026	arXiv:2609.28003	Auto	None	C2	Oct 26
S3	GateMem multi-principal memory-governance benchm	2	2026	arXiv:2606.18829	n/a	n/a	n/a	Oct 26
S3	Gemini Code Assist Memory	4	2025	cloud.google.com				May 26
S3	Generative Agents	3	2023	arXiv:2304.03442				May 26
S3	Governed Collaborative Memory (Cuadros et al.)	5	2026	arXiv:2605.04264				May 26
S3	HippoRAG	2	2024	arXiv				May 26
S3	Honest Lying mitigation / programmatic failure s	4	2026	arXiv:2605.29463	Auto	None	C1/C2	Oct 26
S3	KV-cache rollback consistency (Zhang & Yang)	2	2026	arXiv:2608.15939	n/a	n/a	n/a	Oct 26
S3	LangChain Primitives (legacy)	1	2023	langchain.com				May 26
S3	LangGraph Long-Term Memory	3	2024	langchain.com				May 26
S3	LangGraph Store / LangMem (2026 refresh)	3	2026	docs.langchain.com/oss/pyth	Novel	None	n/a	Oct 26
S3	Learning on the Job	4	2026	arXiv:2607.22157	Auto+Human	None	C2,C3	Oct 26
S3	Letta MemFS + Context Repositories + Dreaming (2	4	2026	docs.letta.com/letta-code/me	Auto	Auto	C3	Oct 26
S3	Mem0	3	2025	arXiv				May 26
S3	Mem0 Platform (2026 refresh: history, feedback,	3	2026	docs.mem0.ai/api-reference/m	Auto	None	C3	Oct 26
S3	MemAudit post-hoc poisoned-memory audit (Tan et	3	2026	arXiv:2605.23723	Auto	None	C2	Oct 26
S3	MemClaw / Governed Shared Memory (Margalit et al	3	2026	arXiv:2606.24535	Auto	None	C3	Oct 26
S3	MemGPT / Letta	3	2023	arXiv:2310.08560				May 26
S3	MemGuard persisted verifier signals (Wang et al.	4	2026	arXiv:2608.21867	Auto-gated	None	C3	Oct 26
S3	MemoryBank	3	2023	Zhong et al.				May 26
S3	Mi-Memory lifecycle framework (Liu et al.; Xiaom	3	2026	arXiv:2607.18975	Auto	None	C3	Oct 26
S3	MIRIX	3	2025	arXiv				May 26
S3	MPBench memory-poisoning study (Dash et al.)	2	2026	arXiv:2606.04329	n/a	n/a	n/a	Oct 26
S3	NanoResearch Memory Module	3	2026	arXiv:2605.10813				May 26
S3	PPMF provenance-preserving memory firewall (Xu e	3	2026	arXiv:2607.29167	Auto	None	C1	Oct 26
S3	REALM retrieval-driven memory reconsolidation (S	4	2026	arXiv:2609.16053	Auto	None	C2/C3	Oct 26
S3	Recuris recursive experiential-working memory (Y	4	2026	arXiv:2608.24876	Auto-gated	None	C2	Oct 26
S3	Reflexion	4	2023	arXiv:2303.11366				May 26
S3	Revocation guard / Revoked but Still Authoritati	3	2026	arXiv:2609.08258	Human	None	C1	Oct 26
S3	Safe to Resume? checkpoint-rollback continuity (	2	2026	arXiv:2608.29381	n/a	n/a	n/a	Oct 26
S3	ShareMem	4	2026	arXiv:2609.32511	Auto	Auto-parti	C3	Oct 26
S3	SIx Harness / Memory as Infrastructure (Helwig)	3	2026	arXiv:2609.05510	Auto	None	C2	Oct 26
S3	SOAR Cognitive Architecture	4	classical	Newell/Laird				May 26
S3	SSGM stability-and-safety governed memory (Lam e	2	2026	arXiv:2603.11768	None	None	n/a	Oct 26
S3	Stale constraints in inherited memory (Nakayashi	2	2026	arXiv:2608.25553	n/a	n/a	n/a	Oct 26

Sub.	System	Score	Year	Source	Loop 1	Loop 2	Class	Added
S3	TMA-NM origin-bound memory authority (Louck)	3	2026	arXiv:2606.24322	Auto	None	C1	Oct 26
S3	TOKI bitemporal operator algebra (Wang)	3	2026	arXiv:2606.06240	Auto or Hu	None	C3	Oct 26
S3	Trust-scored memory sanitization (Devarangadi Su	2	2026	arXiv:2601.05504	Auto gate	None	n/a	Oct 26
S3	Typed provenance + assertion guardrails / Stored	3	2026	arXiv:2609.02127	Auto	None	C1	Oct 26
S3	Utility Under Attack (Karunanidhi)	2	2026	arXiv:2608.21230	n/a	n/a	n/a	Oct 26
S3	VerificAgent human-verified frozen memory (Nguye	3	2025	arXiv:2506.02539	Human	None	C3	Oct 26
S3	VerMem verifiable memory management (Sun et al.)	4	2026	arXiv:2608.03137	Auto-gated	None	C2/C3	Oct 26
S3	Voyager Skill Library	4	2023	arXiv:2305.16291				May 26
S3	Zep / Graphiti	3	2025	arXiv				May 26
S3	Zep / Graphiti bitemporal facts (2026 refresh)	3	2026	help.getzep.com/facts; help.	Auto	None	C3	Oct 26
S4	Anthropic Tool Use API	4	2024	docs.anthropic.com				May 26
S4	Browser Tools (Playwright/Computer Use)	3	2024	anthropic.com				May 26
S4	Claude Code Tool API	4	2024	docs.anthropic.com				May 26
S4	Code Interpreter	3	2023	openai.com				May 26
S4	Cursor MCP Integration	4	2024	docs.cursor.com				May 26
S4	Devin / SWE-agent Tool Harness	4	2024	cognition.ai				May 26
S4	Gemini Function Calling	3	2024	ai.google.dev				May 26
S4	GPT Actions	3	2023	platform.openai.com				May 26
S4	Harvey (Legal Vertical)	5	2024	harvey.ai				May 26
S4	Hippocratic AI	5	2024	hippocraticai.com				May 26
S4	LangChain Tools Abstraction	3	2023	langchain.com				May 26
S4	LangGraph Tool Nodes	4	2024	langchain.com				May 26
S4	LlamaIndex Tool Ecosystem	4	2023	llamaindex.ai				May 26
S4	Model Context Protocol (MCP)	5	2024	modelcontextprotocol.io				May 26
S4	OpenAI Assistants API	3	2023	platform.openai.com				May 26
S4	OpenAI Function Calling	3	2023	platform.openai.com				May 26
S4	Sierra	4	2024	sierra.ai				May 26
S4	SkillSmith	4	2026	arXiv:2606.01314	Auto	None	C1	Oct 26
S4	Toolformer	2	2023	arXiv:2302.04761				May 26
S4	ToolLLM / ToolBench	3	2023	arXiv:2307.16789				May 26
S5	AgentConductor	0	2026	arXiv:2602.17100	None	None	C2	Oct 26
S5	AgentGrad	4	2026	arXiv:2609.08572	Auto	None	C2	Oct 26
S5	AgentVerse	2	2023	arXiv:2308.10848				May 26
S5	AutoGen / MagenticOne	3	2024	arXiv:2308.08155				May 26
S5	AWS Bedrock Multi-Agent	2	2024	aws.amazon.com				May 26
S5	Bilevel Autoresearch	4	2026	arXiv:2603.23420	Auto	None	C1	Oct 26
S5	CAMEL	1	2023	arXiv				May 26
S5	ChatDev	2	2023	arXiv:2307.07924				May 26
S5	ContextEvo	4	2026	arXiv:2609.34649	Auto	None	C2	Oct 26
S5	Control System for Frozen Agents (Paul)	4	2026	arXiv:2607.25415	Auto	None	C2	Oct 26
S5	CrewAI	3	2025	docs.crewai.com				May 26
S5	Evo2Team	4	2026	arXiv:2609.34135	Auto	Auto	C2	Oct 26
S5	Growing Harness	4	2026	arXiv:2609.26760	Auto	None	C2	Oct 26
S5	HARBOR	4	2026	arXiv:2604.20938	Auto	None	C2	Oct 26
S5	HarnessForge	4	2026	arXiv:2606.01779	Auto	None	C2	Oct 26
S5	Inspect AI	2	2024	ukgovernmentbeis.github.io				May 26
S5	Institutional MAS architectures (Fei et al.)	1	2026	arXiv:2604.27691	None	None	C2	Oct 26
S5	LangGraph	3	2024	langchain.com				May 26
S5	Letta	3	2025	letta.com				May 26
S5	LlamaIndex Workflows	2	2024	llamaindex.ai				May 26
S5	Memento-Skills	4	2026	arXiv				May 26
S5	MetaGPT	2	2023	arXiv:2308.00352				May 26
S5	Microsoft Semantic Kernel	2	2024	learn.microsoft.com				May 26
S5	Multi-Agent Evolve (MAE)	4	2025	arXiv				May 26
S5	OpenAI AgentKit	4	2025	openai.com				May 26
S5	OpenAI Swarm	1	2024	openai.com				May 26
S5	OpenHands / CodeAct	3	2024	arXiv:2407.16741				May 26
S5	OptiMAS	4	2026	arXiv:2608.21918	Auto	None	C2	Oct 26
S5	PluginRSI	4	2026	arXiv:2609.32423	Auto	None	C2	Oct 26
S5	Pydantic AI	2	2024	pydantic.dev				May 26
S5	Recursive Harness Self-Improvement (RHI)	4	2026	arXiv:2607.15524	Auto	None	C3	Oct 26
S5	Skill-MAS (meta-skill)	4	2026	arXiv:2606.18837	Auto	None	C2	Oct 26
S5	TACO	4	2026	arXiv:2604.19572	Auto	None	C2	Oct 26
S5	Task-CoEvolve	4	2026	arXiv:2608.20169	Auto	None	C2	Oct 26
S5	TOCOMAS	4	2026	arXiv:2609.37953	Auto	None	C2	Oct 26
S5	Vertex AI Agent Builder	4	2024	cloud.google.com				May 26
S5	Workflow-Designing Agents (WDA)	4	2026	arXiv:2609.32458	Auto	None	C2	Oct 26
S6	Agenta	3	2024	agenta.ai				May 26

Sub.	System	Score	Year	Source	Loop 1	Loop 2	Class	Added
S6	Agentic Configuration Management ACM (Quessada-V	3	2026	arXiv:2608.11166	None	Human	n/a	Oct 26
S6	AGENTS.md / AAIF	5	2025	agents.md				May 26
S6	Anthropic skills repo	4	2025	github.com/anthropics/skills				May 26
S6	Azure API Center MCP registry	3	2026	learn.microsoft.com (ms.date	Human	Human	-	Oct 26
S6	Braintrust	5	2024	braintrust.dev				May 26
S6	Cleric.ai	2	2025	cleric.ai				May 26
S6	Competence Not Accuracy	2	2026	arXiv:2608.18719	None	None	-	Oct 26
S6	Constitutional AI	2	2022	Anthropic				May 26
S6	Cursor Rules	3	2024	cursor.com				May 26
S6	Decagon	2	2026	decagon.ai				May 26
S6	DSPy compile	3	2024	dspy.ai				May 26
S6	EvolveNet	4	2026	arXiv:2608.04968	Auto	Auto	C2	Oct 26
S6	EvoUndo	4	2026	arXiv:2608.28363	Auto	None	C1	Oct 26
S6	From Prompts to Contracts	2	2026	arXiv:2607.08028	None	None	C1	Oct 26
S6	Future AGI prompt lifecycle	4	2026	futureagi.com/blog (2026-03-	Auto	Auto	C3	Oct 26
S6	GitHub Copilot enterprise	4	2026	github.com				May 26
S6	Governance conversion (Davis et al. case study)	3	2026	arXiv:2607.01087	Human	None	C2	Oct 26
S6	Governed Capability Evolution (Qin et al.)	3	2026	arXiv:2604.08059	None	Auto	C2	Oct 26
S6	HackProbe	3	2026	arXiv:2609.04665	Auto	None	C1	Oct 26
S6	Inspect AI	3	2024	ukgovernmentbeis.github.io				May 26
S6	LangFuse	3	2024	langfuse.com				May 26
S6	LangSmith Hub	5	2024	smith.langchain.com				May 26
S6	MCP Gateway & Registry (agentic-community)	3	2026	github.com/agentic-community	Human	Human	C1 (s	Oct 26
S6	NanoResearch	2	2026	arXiv:2605.10813				May 26
S6	Official MCP Registry	2	2026	modelcontextprotocol.io/regi	Human	None	-	Oct 26
S6	OpenAI AgentKit Evals	3	2025	openai.com				May 26
S6	OPRO	2	2023	arXiv				May 26
S6	Organizational Control Layer (Shi et al.)	2	2026	arXiv:2606.04306	None	None	n/a	Oct 26
S6	Registry-Governed Agent Lifecycle / EDDOps on AW	3	2026	arXiv:2607.00345	None	Auto+Human	C3	Oct 26
S6	Reward-Free Pairwise Validator	4	2026	arXiv:2607.14408	Auto	None	C3	Oct 26
S6	Runtime Skill Audit	2	2026	arXiv:2606.11671	None	None	-	Oct 26
S6	Runtime-Independent Persistent Agents (Enoch)	3	2026	arXiv:2609.00546	Human	Human	C1	Oct 26
S6	SafeEvolve	4	2026	arXiv:2608.12851	Auto	None	C3,C2	Oct 26
S6	SAGE statistical acceptance gate (≠ SAGE 2512.17	4	2026	arXiv:2609.36043	Auto	None	C2	Oct 26
S6	Self-Refine	1	2023	Madaan et al.				May 26
S6	Sierra Ghostwriter	2	2026	sierra.ai				May 26
S6	SkillCoach	3	2026	arXiv:2607.01874	Auto	None	C3,C1	Oct 26
S6	SkillProbe	2	2026	arXiv:2603.21019	None	None	-	Oct 26
S6	SkillSpec (intent-masked correctness)	2	2026	arXiv:2609.06052	None	None	-	Oct 26
S6	TextGrad	2	2024	arXiv				May 26
S6	TrueFoundry MCP Gateway	3	2026	truefoundry.com/docs (mcp-ga	Human	Human	-	Oct 26
S6	Vellum	5	2024	vellum.ai				May 26
S6	Verifier-as-Gatekeeper (VaG)	4	2026	arXiv:2608.05810	Auto	None	C2	Oct 26
S6	W&B Weave (Prompts)	3	2024	weave-docs.wandb.ai				May 26

S2. Data appendix

Corpus size. The survey scores 130 unique LLM systems across the six substrates (140 substrate-level rows in `supplementary/p3_master_scores.csv`). Each system may be scored on multiple substrates; the per-substrate row count therefore exceeds the unique-system count.

Core / contrast split. 99 of the 130 unique systems satisfy the patch-local discriminator (score ≥ 3 on at least one substrate) and form the core corpus. The remaining 31 systems form the contrast corpus: commonly described as agentic or self-improving in public discourse but failing at least one discriminator. Appendix D enumerates representative contrast systems.

Inclusion criteria. Any system that (a) wraps a foundation model with at least one persistent artifact updated from deployment signal, or (b) is named in the public discourse as a "self-improving," "scaffolded," or "agentic" system. The contrast set (Self-Refine, ReAct, TextGrad, OPRO, engineer-authored convention

files; Appendix D.1) is included so the discriminator has something to mark against; Appendix D.2 separately lists core-corpus systems that fail the composite two-loop criterion.

Exclusion criteria. Systems whose only update mechanism is weight-level fine-tuning or RLHF: these are model-side, not scaffold-side, and belong to the self-evolution literature this paper differentiates from. Systems with no public evidence of deployment (white papers without code, demos, or production claim).

Evidence tiers. Each system carries one or more of: (T1) peer-reviewed publication; (T2) arXiv preprint; (T3) production claim with metrics; (T4) open-source artifact; (T5) industry-blog or interview-level documentation. T1/T4 carry highest weight in cluster analysis; T3 is admissible where the claim is concrete (specific benchmark, named integration target); T5 is admissible only when corroborated by T1–T4 elsewhere. Tier flags are propagated to cluster claims through the per-row evidence column in Appendix A; T3 production claims are not pooled with T1/T2 evidence when computing cluster-level statistics.

Evidence-tier distribution. Of the core corpus, approximately one quarter carry T1 evidence, roughly half carry T2 evidence, and the remainder carry T3 or T4 evidence as the primary tier.

Survey window and dating. January 2023 – May 2026 for systematic corpus assembly and scoring, extended by a targeted literature refresh through September 2026 that added the systems marked § in Appendix A (FederatedSkill, SkillClaw, ChronoMem, SkillsVote, HARNESSEVO) and the supporting references of §§8–10. The refresh searched for work that would fill or narrow the §9 corner, not for corpus completeness, so the per-substrate statistics of §6 are computed on the May-2026 corpus. Preprint-only systems carry T2 evidence and are flagged in Appendix A. A fraction of T2 work is recent enough that ablations may yet be revised; cluster-level claims are robust to dropping any single T2 system.

Scoring discipline. Each system received a substrate-by-substrate 0–5 score against the rubric of Section V of the main text and an integrated patch-local-adaptation score (PP) in {0,1,2,3,3.5,4,4.5,5}. Half-scores were used sparingly when one criterion was clearly met and another partially. Ambiguous cases were resolved conservatively (downward). Scoring is single-rater; Appendix A and Appendix D together expose representative per-system scores and exclusion reasons so readers can audit ambiguous classifications independently.

Composite two-loop criterion (sensitivity analysis). The strict count of fourteen research systems satisfying the composite-two-loop criterion changes under two natural relaxations. Relaxation (a) counts any of versioning, lineage, or rollback as Loop-2 governance; the count rises to approximately eighteen by re-admitting MAE, MetaGen, MetaReflection, and CLIN. Relaxation (b) admits session signal that updates optimizer state without persisting an artifact distinct from optimizer state; the count rises to approximately sixteen by re-admitting DSPy and TextGrad. The qualitative findings (Auto×Auto well-populated; [Human × Human] and [Human × Auto] empty in the surveyed corpus; governance-first cluster in [None × Auto]; the cross-tenant governed-optimization cell empty) survive both relaxations.

S3. The mechanism stack and continual-learning notes

S3.1 The mechanism stack

The abstract update rule of Section III expands into six operations that every patch-local system must perform, whether explicitly engineered or accidentally emergent.

1. **Residual capture.** Log failures, corrections, rejected outputs, user edits, eval regressions, tool-call failures. The Loop-1 signal source. Without it, z_t has no provenance.
2. **Mode discovery and credit assignment.** Cluster repeated failures into patch-specific error modes (the *Architecture of Errors* catalog Arbuzov et al. [2026], instantiated locally per patch) and attribute each mode to the scaffold coordinate responsible for it (E.6).
3. **Delta synthesis.** Convert a recurring mode into a candidate scaffold edit: an instruction rewrite, a new skill, a memory correction, a tool binding, an orchestration change, an eval case. This produces z_t.

4. **Delta validation.** Estimate whether the edit reduces L_D without collateral damage on adjacent inputs. The gate G_t is Auto-gated when the check is an algorithmic criterion, Human-gated when it is reviewer judgment.
5. **Promotion control.** Decide whether the accepted edit stays in the local fork or climbs the scope hierarchy: project, stack, organization, cross-tenant. Loop 2.
6. **Drift and bloat control.** Expire stale rules, prune conflicting memories, roll back regressed skills, detect over-fitting. Without this, accumulated edits produce the scaffold-level analog of plasticity loss.

Each operation is the locus of a separate engineering discipline. Surveyed systems implement them partially and unevenly: research full-auto systems excel at (3) and (4); production governance leaders at (5); industry hybrids at most of (1)–(5) but rarely (6). Implementation patterns are deferred to follow-on practitioner work.

S3.2 Loop 1 and Loop 2 as descent steps of different radius

Loop 1 proposes and validates local descent steps on a single L_D. Loop 2 expands the update radius. It selects, merges, and promotes accepted local edits so that they apply to a larger averaging set of patch losses. USER-level acceptance descends on a single L_D. PROJECT promotion descends on a project-averaged loss. STACK promotion widens the average further. CORE or cross-tenant promotion descends on a federated average. The two-loop design space is therefore not just a taxonomy of engineering patterns: it is a taxonomy of *which loss each surveyed system is descending on*, gated by which estimator.

S3.3 Overfitting in scaffold adaptation

In weight training, overfitting is usually a defect. In scaffold adaptation, local overfitting is partly the point. Frontier weights are trained to preserve cross-patch generality; they must smooth over local conventions and rare workflow-specific failures. A deployment patch needs the opposite: selective specialization to local residuals. The scaffold is therefore an intentionally overfittable layer, but one whose overfitting is scoped, inspectable, versioned, gated, and reversible. A scaffold that improves one repository, hospital workflow, or legal diligence process may degrade performance elsewhere. That is not a contradiction; it is patch adaptation. The mistake is promoting that edit beyond its evidence radius.

In non-stationary patches ($D = D_t$), the same descent mechanism tracks a moving optimum rather than converging once. This is why recency weighting, replay, pruning, expiry, and rollback are not optional hygiene; they are the scaffold-level analogs of continual-learning machinery.

S4. Contrast set, matrix membership, and closest approaches (May 2026 corpus)

The October 2026 additions are placed in the cells of Section VII of the main text and listed in S1; the per-system discussion below is the May 2026 audit on which the main text's counts build.

S4.1 Contrast set

Appendix D serves two audits. **D.1** lists representative members of the contrast corpus (31 systems that fail the patch-local discriminator, score < 3 on every substrate), so the discriminator has something to mark against. **D.2** lists core-corpus systems (score ≥ 3 on at least one substrate) that nonetheless fail the stricter composite two-loop criterion of Section V, so the asterisked entries of the Section VII matrix are auditable. Systems named in the text but not scored as rows in `supplementary/p3_master_scores.csv` (ReAct, MetaGen, MetaReflection, CLIN, ACE) are classified from their papers and marked with a dagger.

S4.1.1 Contrast corpus: fails the patch-local discriminator

System	Year	Discriminator failure	Brief reason
Self-Refine Madaan et al. [2023]	2023	No persistent artifact	In-context iteration only
ReAct Yao et al. [2023]†	2023	No persistent artifact	Thought-act-observe loop is intra-session
TextGrad Yuksekgonul et al. [2024]	2024	No artifact distinct from optimizer state	Textual-gradient optimization; SkillOpt [Yang et al. [2026b]; Section III.5; Appendix G] adds the persistent `best_skill.md`, held-out validation gate, rejected-edit memory, and textual learning-rate schedule that TextGrad lacks.
OPRO, Toolformer	2023	No persistent deployment artifact	Optimization or tool use inside a single training or inference run
Aider conventions, Lovable knowledge	2024	No feedback loop	Engineer-authored convention files
AGENTS.md (as a standard)	2025	No feedback loop	Cross-vendor artifact format; adoption measured, adaptation absent (Section VI)
LangChain Skills, HuggingFace Agent Skills, CrewAI Skills	2025–2026	No feedback loop	Curated skill packs without failure-triggered update
Tran & Kiela Tran and Kiela [2026]	2026	Not a system	Single-agent baselines match or beat multi-agent under matched thinking-token budgets; constrains S5 claims (see Section VI)

S4.1.2 Core corpus, fails the composite two-loop criterion

System	Year	Composite-criterion failure	Brief reason
Reflexion Shinn et al. [2023]	2023	No Loop 2	Per-agent episodic memory without promotion
Generative Agents	2023	No Loop 2	Per-agent reflection + importance memory
ExpeL	2023	Loop boundary blurred	Experience bank built from training tasks
Mem0 (vanilla)	2024	No Loop 2	Per-user memory; no cross-context promotion
Zep	2024	No Loop 2	Per-agent/session knowledge graph

System	Year	Composite-criterion failure	Brief reason
ChatGPT Memory	2024	No Loop 2	Per-user persistence
Claude Memory	2025	No Loop 2	Per-conversation-thread persistence
MemGPT	2023	No Loop 2	Virtual-context management per agent
DSPy	2023	No artifact distinct from optimizer state	Prompt-template optimization; SkillOpt [Yang et al. [2026b]; Section III.5; Appendix G] realizes the artifact-bearing form.
MetaGen†	2026	No Loop 2; no cross-session persistence	Role pool dynamic within single inference call
MAE (Multi-Agent Evolve)	2025	No Loop 2	Co-evolution within Proposer/Solver/Judge triplet
MetaReflection†	2024	No Loop 2	Per-agent semantic memory
CLIN†	2023	No cross-instance promotion	Causal abstractions per-agent-per-environment
LangGraph, AutoGen, CrewAI	2023–2024	Framework only	Orchestration frameworks; not deployed systems
W&B Weave, Agenta, Langfuse	2024	No Loop 1	Observability / eval-as-code
Mobile-Agent-E, Cradle, OS-Copilot	2024–2025	No Loop 2	Self-evolving but per-session/per-deployment
Agent S, Agent S2	2024–2025	No Loop 2	Computer-use agentic framework
ACE Zhang et al. [2025]†	2025	No Loop 2 in surveyed form	Single-context evolving playbook
Cursor Rules (vanilla)	2024	No Loop 2	Per-project artifacts
GitHub Copilot custom instructions, Windsurf, Continue.dev, Zed AI rules	2024	No Loop 1	Engineer-authored at repo base branch
Replit `replit.md`	2025	Loop 2 absent	Self-writing scaffold but per-workspace

The entries above cover the main exclusion families (in-context-only; no feedback loop; not a system) and the main composite failures (no Loop 2; framework-only; engineer-authored without failure trigger; optimizer-state-only); finer-grained per-system breakdowns are deferred to follow-on practitioner work.

S4.2 Survey detail

This appendix carries the cell-by-cell discussion of the two-loop matrix from Section VII, the closest-approaches enumeration that supports Section IX, and the deployment-dependence argument. It supplies the verification material for §Section VII–9.

S4.2.1 Two-loop matrix: cell-by-cell discussion

The named membership of the Section VII matrix (Figure 3):

	Loop-2 Auto-gated	**Loop-2 Human-gated**	**Loop-2 None**
Loop-1 Auto-gated	NanoResearch, SkillOpt, SkillRL, AlphaEvolve, ADAS, DGM, Voyager, SAGE, EvolveR, CoEvoSkills, AutoAgent, AutoManual	SkillForge, CASCADE, Sierra Agent OS 2.0†, Cognition Devin†	MAE*, MetaGen*, MetaReflection*, CLIN*, DSPy*, TextGrad*
Loop-1 Human-gated	[**empty by engineering logic**]	[**empty in surveyed corpus**]	*(none)*
Loop-1 None	ExpeL (weak), Braintrust*, LangSmith Hub*, Vellum*	Anthropic skills repository*	Self-Refine*, ReAct*, Mem0*

Asterisks fail at least one composite-two-loop criterion; included for contrast. † Industry exemplar evaluated using vendor documentation as primary evidence. DSPy and TextGrad sit in [Auto × None] because their Loop 1 is automated but persists only optimizer state (Appendix B, relaxation (b)). The matrix is restricted to the 22 systems of the strict composite audit plus SkillOpt; Appendix A also assigns Loop-1 / Loop-2 entries to systems outside that audit (COSPLAY, Cuadros et al., PACE, CORAL, GEPA, Meta-Harness, MOSS, RCL, marked ‡ there, and the September-2026 refresh additions marked §), but those entries are read from the papers and were not re-audited under the composite criterion, so they are neither placed in the matrix nor counted in Section V.

[**Auto × Auto**] is most populated (twelve systems). Every member has a machine-evaluated decision criterion on both loops. Velocity is the cluster's defining property; the lack of cross-organizational governance is its uniform weakness.

[**Auto × Human**] is the production-viable cell: Loop 1 fires at machine speed under an algorithmic criterion, Loop 2 requires human approval before propagation. The pattern reconciles automation velocity with reviewer accountability. SkillForge's LLM-judge + reviewer pipeline is the architectural exemplar. CASCADE, Sierra Agent OS 2.0, and Devin instantiate variants; the industry exemplars (Agent OS 2.0, Devin) are marked with † because vendor documentation rather than peer-reviewed metrics is the primary evidence.

[**None × Auto-gated**] holds the governance-first prompts-as-code systems (Braintrust, LangSmith Hub, Vellum). They have a real Loop 2 (eval-gated CI blocks promotion on regression), but no Loop 1 in the patch-local sense, because the candidate artifacts are authored by engineers at their desks rather than triggered by session feedback. The asterisks mark this: governance without inner-loop adaptation. The cell is informative because it shows what survives when only the outer loop is automated.

[**Human × Auto**] is empty by engineering logic: if humans author candidates manually, automated promotion offers little leverage. A speculative occupant would be "expert curation + auto-distribution" (expert-authored skills automatically gated through a held-out eval before shipping); no published example exists.

[**Human × Human**] is empty in the surveyed corpus. A candidate occupant would be a reviewer-authored-and-reviewer-promoted artifact pipeline targeting solo and small-team workflows where automation cost exceeds review latency. No public example was found in the survey window.

[**Human × None**] is structurally improbable: a system requiring human authorship on Loop 1 but with no Loop 2 collapses to "engineer-edits-a-config" and is not patch-local in the sense of Section III.

DGM dual-mode note. DGM operates in both [Auto × Auto] (primary archive-based evolution) and [Auto × Human] (sandboxed code-modification review) modes depending on deployment configuration; we list it once in the dominant cell.

S4.2.2 Closest approaches to governed cross-tenant scaffold optimization

The four properties of governed cross-tenant scaffold optimization are (a) auto-gated Loop 1, (b) multi-tenant cross-org promotion, (c) versioned lineage with RBAC, (d) rollback. Closest approaches partition the requirements:

- **AlphaEvolve** has (a) and partial (b) within a single organization; no cross-org promotion mechanism.
- **NanoResearch** has (a) at the research level; no enterprise governance layer described.
- **SkillOpt** has (a) fully and partial (c) via the exported `best_skill.md` artifact plus protected slow-update field; reports positive cross-model, cross-harness, and cross-benchmark transfer as empirical evidence-radius measurement; does not implement (b) cross-organizational promotion or (d) explicit rollback. The Appendix G walkthrough maps every SkillOpt design decision to its Section III counterpart.
- **CORAL** Qu et al. [2026] has partial (a) via heartbeat-gated promotion and is the only system in the corpus that implements asynchronous multi-agent Loop 2 within one organization through shared persistent memory; no (b) cross-organizational scope, no formal (c) RBAC, no (d) rollback.
- **PACE** Ling et al. [2026] has (a) at the small-model production timescale and a fast/slow two-timescale Loop 1 + Loop 2 that consolidates across one deployment's own episodes rather than across deployments; no (b), no formal (c) or (d). PACE is the closest demonstration that the two-loop architecture is implementable under small-frozen-model constraints.
- **Meta-Harness** Lee et al. [2026] has (a) over the harness substrate (S5) with a filesystem-of-candidates credit-assignment record that functions as the accumulated history H_t of Section III; no Loop-2 cross-organizational promotion, no RBAC, no rollback.
- **SkillForge** has (a) and (b) within one enterprise, partial (c) via VFS commits; (d) is undocumented in the public report.
- **Sierra Agent OS 2.0** has (b) and partial (a); the GitHub-style Workspaces provide some lineage but not RBAC-with-rollback in the strict sense.
- **FederatedSkill** Yang et al. [2026a] has (a) and the strongest partial (b) in the corpus: heterogeneous clients (backbone-plus-harness profiles under one federation operator) exchange semantic skill diffs, and a server-side evolution agent applies, adapts, or rejects each patch per client; partial (c) via an auditable decision log recording source patch, reward, and justification per modified path; no RBAC, no (d).
- **SkillClaw** Ma et al. [2026] has partial (a) via a nightly validation step and partial (b) across users of one shared, synchronized skill repository; no (c) or (d) reported.
- **ChronoMem** Su et al. [2026] has (d) and the versioning half of (c) for the memory substrate alone (whole-memory snapshots, version histories, natural-language rollback inside Google's Agent Development Kit); no Loop 1, no (b), no RBAC.
- **Vellum / Braintrust / LangSmith Hub** have (c) and (d) explicitly but no (a) or (b) in the strict sense; these are governance-first systems by construction.

No surveyed system combines (b), (c), and (d) under any gate type. The cross-organizational governance triad is itself unfilled, independently of whether Loop 1 is auto-gated.

S4.2.3 Deployment-dependence of the missing architecture

Whether the corner *should* be filled is deployment-dependent. In safety-critical domains (medical decision support, legal advice, financial advice), reviewer-judgment gates may be a design feature rather than a defect, and the human latency is the point. The narrower survey claim is what matters: no public system currently combines automated local scaffold evolution with governed cross-organization promotion, versioned lineage with RBAC, and rollback. Naming the corner converts a vague gap into a specific four-property audit criterion that any future system can be measured against. The shortest demonstrable path appears to build on AlphaEvolve's cascade evaluator and MAP-Elites archive, extending federation from within-DeepMind multi-target to cross-customer multi-tenant. The architectural surface, not the timeline, is what this paper claims.

S5. The convergent research families, system by system

This section carries the May–September 2026 walkthroughs; the October 2026 additions to each family are described in Section VIII of the main text and scored in S1.

This appendix expands Section VIII along two axes. G.1 indexes the landscape; G.2–G.3 walk SkillOpt mechanism-by-mechanism; G.4–G.8 walk each convergent research family, stating its Section III mapping explicitly and naming what it leaves open; G.9 synthesizes four claims supported by the families as a whole; G.10–G.11 carry the Section IX-property audit of SkillOpt and its published cost-per-point figure.

S5.1 The convergent landscape

SkillOpt Yang et al. [2026b] is the cleanest current instance of artifact-layer descent in our survey, but it sits alongside at least eighteen independent research threads in 2025–2026 that converge on the same disciplined gradient-like update mechanism for the scaffold layer. The convergence is striking because it happens *from different starting points*: GEPA and Tiwari et al. start from prompt-optimization and continual-learning respectively; RCL starts from cognitive-agent learning; Meta-Harness, AHE, MOSS, CANTANTE, and Ong et al. start from harness engineering; MOCHA and Ratchet start from skill-deployment constraints; ICT, Combee, CORAL, and PACE start from cross-task / multi-agent / production-deployment angles. Each system instantiates a different subset of Section III's five conditions, and each exposes a different gap.

The landscape table of Section VIII indexes the families by Section III mechanism and by what each leaves open; the subsections below walk each family in detail. By substrate (Section V), GEPA and RCL operate on the prompt and context layer (S1–S2); the harness family on orchestration (S5), with MOSS extending to source code (S2+S5); MOCHA and Ratchet on skills under deployment constraints (S2); the transfer family on integrated memory-plus-orchestration stacks (S3+S5); and SkillOpt on an S2-led integrated stack.

Read against Section IX, no member of any family combines (a) auto-gated Loop 1 with (b) cross-organizational Loop 2, (c) versioned lineage with RBAC, and (d) rollback. The Section III picture is realized in fragments across the families; the Section IX gap is confirmed by every member.

S5.2 SkillOpt: full correspondence

The table below extends the Section VIII compact correspondence with the Section III role of each component and an algorithm-level pointer into SkillOpt's published procedure (their Algorithm 1).

Section III quantity	SkillOpt instantiation	Section III role explained	Algorithm pointer
θ_s (scaffold)	`best_skill.md` — a 379–1,995-token markdown document	the trainable external state of the frozen target model; persistent and exported as the deployment artifact	s_{best} (Algorithm 1, lines 1, 22, 37)
L_D (patch loss)	benchmark-native hard score / exact-match on a held-out test split	scalar quality signal that the optimizer is implicitly minimizing	$\textsc{Evaluate}(M, h, s, D)$ (lines 2, 17, 37)
Q_t (proposal)	optimizer-model reflection over failure and success minibatches; failures and successes analyzed separately, then merged with priority on failure fixes	proposal distribution conditioned on accumulated rollout evidence and current scaffold	optimizer calls O for failure analysis, success analysis, and merge (lines 9–11)

Section III quantity	SkillOpt instantiation	Section III role explained	Algorithm pointer
z_t (edit)	structured add/delete/replace patch on the skill (or rewrite-from-suggestions in rewrite mode)	the candidate update applied via $\oplus$ to the current scaffold	rank/clip to η_t edits, apply to obtain $\tilde{s}$ (lines 12–13)
η_t (textual learning rate)	bounded per-step edit budget; constant / linear / cosine / autonomous schedules; default cosine with floor $\eta_t = 2$	edit-magnitude controller; the Section III *learning rate* in textual form	rank/clip step (line 12); schedule applied per epoch
G_t (gate)	held-out selection-split accept gate, strictly greater than current selection score	the per-edit gate G_t — accepts only when $\widehat{\Delta}_{z_t} L_D$ improves the selection score	comparison at line 20; cache lookup at lines 14–18
rejected-edit buffer	epoch-local store of failed proposals and the score drop they caused	converts rejected steps into negative feedback for later proposals within the same epoch; zero deployment cost	$\mathcal{B}$ updated at line 26; reset at line 5
slow/meta update	epoch-end longitudinal guidance written to a markup-fenced protected slow-update field, still passed through D_{sel}	second-order consolidation across adjacent epochs; the Section III "step-level edits cannot overwrite the protected slow-update field" decomposition	lines 28–31; gate applied to injected guidance
optimizer-side meta skill	m_{meta} — a prompt-side summary of which edits helped, which failed, and which failures persisted	training-time guidance for Q_t; never shipped with the deployed artifact	lines 33–34 (training-side only)

The mapping is one-to-one at the level of Algorithm 1: every Section III quantity has a named, code-level counterpart, and every SkillOpt control surface maps onto a Section III quantity. SkillOpt is therefore the cleanest published instance we are aware of in which the Section III picture is *mechanism* rather than *metaphor*: all five Section III conditions are instantiated.

S5.3 What SkillOpt does that prior textual-optimization systems do not

SkillOpt's own experiment design compares it against six baselines under an identical protocol (same target model, same held-out test split, same scorer). The differentiation below frames each baseline in Section III vocabulary and identifies the Section III mechanism the baseline lacks. The through-line is the paper's thesis: *the scaffold/evolution layer needs a disciplined gradient-like update mechanism*, and the live distinction across baselines is which Section III conditions each one fails.

- **TextGrad** Yuksekgonul et al. [2024]**.** Backpropagates textual feedback through `tg.Variable` objects in a compound system. In Section III terms: z_t-style edits exist, but θ_s does not — TextGrad has no artifact distinct from optimizer state. There is no η_t schedule, no held-out G_t, and no rejected-edit buffer. SkillOpt's measured advantage over TextGrad on direct chat ranges from +5.9 to +34.0 points across benchmarks at GPT-5.5 scale and is uniformly positive at all seven target-model scales. This matches our Appendix D classification of TextGrad as failing the patch-local discriminator on persistence; SkillOpt closes precisely that gap.

- **GEPA** Agrawal et al. [2025]**.** Reflective prompt evolution with Pareto-style multi-candidate selection. Closer than TextGrad in that it carries multiple candidates and a selection step. In Section III terms: Q_t is real, but the optimization target is a prompt rather than a persistent skill artifact, G_t is not the strictly-greater held-out gate of Section III, and there is no η_t bound — accepted edits can wholesale rewrite. SkillOpt matches or outperforms GEPA in every reported cell of its main results table Yang et al. [2026b]; we do not restate the per-cell deltas here.
- **Trace2Skill** Ni et al. [2026]**.** Distils trajectory-local lessons into reusable skill artifacts. In Section III terms: θ_s exists, but there is no iterative G_t — Trace2Skill mines skills offline rather than running a held-out validation loop that accepts or rejects each candidate. Our Section III mechanism stack therefore loses *delta validation* (operation 4) and the *rejected-edit buffer*. SkillOpt's measured gap over Trace2Skill is +5.0 to +19.0 points across benchmarks at GPT-5.5 direct chat.
- **EvoSkill** Alzubi et al. [2026]**.** Skill-folder evolution under failure analysis, deployed harness-side. The strongest baseline on the Codex and Claude Code harness rows. In Section III terms: θ_s is a folder rather than a single artifact, and credit assignment via failure analysis is genuine, but there is no η_t bound on per-step folder edits and no rejected-edit buffer. SkillOpt beats EvoSkill by +14.0 inside Codex and +3.2 inside Claude Code on the GPT-5.5 five-benchmark average.
- **Human skill** (expert-written, 145–516 tokens per benchmark). In Section III terms: θ_s exists, but Loop 1 does not — the artifact is static at deployment. Provides the no-Loop-1 ceiling.
- **One-shot LLM skill** (generated zero-shot from a benchmark description by GPT-5.5, never updated). In Section III terms: θ_s exists; Loop 1 does not; Q_t fires exactly once with no rollout evidence. The "untrained scaffold" baseline.

ABSTRAL Song et al. [2026] and EvoTest He et al. [2025], referenced in SkillOpt's related work but not measured in their table, operate at the multi-agent design / test-time configuration layer rather than the persistent skill layer; they are out of scope for the artifact-layer-descent comparison.

Synthesizing across all six measured baselines: the live difference is whether the loop has (i) a persistent edit-target distinct from optimizer state (θ_s), (ii) a held-out validation gate on each accept (G_t with $\widehat{\Delta}$ estimator and accept threshold τ_t), (iii) negative-feedback retention (rejected-edit buffer feeding Q_t), and (iv) a bounded learning rate with a schedule (η_t + cosine/linear/constant/autonomous). SkillOpt is the first published system in our survey that instantiates all four simultaneously, and it does so as a single offline training loop whose deployed artifact remains a 379–1,995-token markdown file requiring no inference-time optimizer calls.

A follow-up, SkillOpt-Lite Shen et al. [2026], asks which of these controls are necessary. It reformulates skill optimization as zeroth-order optimization, keeps three principles (file-system-based trajectory exploration, consensus attribute mining that extracts attributes common across heterogeneous rollouts and discards single-sample eccentricities, and an independent held-out validation gate that is strictly disjoint from the training failures), and removes minibatch reflection pooling, the rejected-edit buffer, the slow/meta update, and the textual learning-rate schedule. It reports faster convergence and higher scores than full SkillOpt (LiveMath +8.8 points at GPT-5.5 and +25.4 at GPT-5.4-nano). Read against the four conditions above, this is evidence that (ii) the gate and a consensus form of credit assignment are load-bearing, while (iii) negative-feedback retention and (iv) the bounded learning rate are stabilizers whose value is benchmark-dependent; the finding does not touch the Section IX governance properties.

S5.4 GEPA, Tiwari, and the reflective-evolution family

The immediate predecessor to SkillOpt's optimizer framing is **GEPA** Agrawal et al. [2025]. It shows that natural-language reflection over execution trajectories can provide a richer adaptation signal than sparse scalar rewards for prompt evolution in compound AI systems. GEPA samples trajectories containing reasoning steps, tool calls, and tool outputs; reflects on them to diagnose failures and propose prompt updates; and maintains a Pareto frontier of candidate prompts rather than greedily mutating only the current best. Across six tasks, GEPA outperforms GRPO by 6% on average and by up to 20% while using up to 35× fewer rollouts,

and also outperforms MIPROv2 in the reported aggregate comparison. In Section III vocabulary, GEPA strongly instantiates the proposal side, Q_t, through trajectory-conditioned reflection, and uses Pareto-based candidate selection as a proposal/pruning mechanism. What it does not instantiate is the full SkillOpt-style artifact lifecycle: a bounded edit budget on a reusable skill document, a strictly-greater held-out gate on each accepted skill update, a rejected-edit buffer used as negative feedback, and an exported deployment artifact with governance hooks.

Tiwari et al. [2026] supply the broader theoretical framing from the optimization side, arguing explicitly that restricting learning to *either* in-context (prompt optimization) *or* in-weights (RL / fine-tuning) is unnecessary and costly. In-context adaptation is cheap and rapid but capacity-bounded; in-weights adaptation is expressive but induces catastrophic forgetting and loss of plasticity Dohare et al. [2024]. The scaffold layer of Section II is precisely the substrate that absorbs what in-context adaptation can carry cheaply, leaving the weights free to generalize. Tiwari et al. supply the two-timescale motivation from the optimization side; Section II supplies the matching motivation from the production-reliability side. The two framings are independent and mutually reinforcing.

S5.5 Reflective Context Learning: a parallel formalization

The most direct independent convergence on the Section III formalization comes from **Reflective Context Learning** [RCL; Vassilyev et al. [2026]]. Their opening observation is that "the fundamental problems of learning, including credit assignment, overfitting, forgetting, local optima, and high-variance learning signals, persist whether the learned object lies in parameter space or context space," and that current methods are "fragmented and ad hoc" precisely because they address these challenges one at a time rather than as a unified optimization problem over the context. The parallel to Section III's argument — that scaffold maintenance needs the discipline of weight-space optimization applied to an external artifact layer — is exact and arrived at independently.

RCL maps onto Section III as follows. The *learned context* is θ_s. The *reflection signal* is a form of credit assignment identifying which context coordinate caused a failure (Appendix E.1, operation 2). The *generalization penalty* that RCL introduces to prevent narrow sample-specific rules is structurally the complexity penalty $\Omega(z)$ of the gate condition. Where RCL and Section III differ is in Loop-2 scope: RCL is framed around single-agent cross-task generalization, while Section III's Loop 2 is cross-deployment, cross-tenant, and cross-organizational promotion. The convergence is nonetheless significant: two research threads starting from different motivations (production reliability in Section III, cognitive agent learning in RCL) arrive at the same formal observation that the classical learning problems do not disappear when the learned object moves outside the weights.

S5.6 The harness-optimization cluster

A second convergent family focuses on the *harness* — the code and configuration layer that wraps a frozen model — rather than on skill documents specifically. This family independently identifies credit assignment as the central unsolved problem of scaffold evolution, and its engineering solutions directly instantiate Appendix E.1's operation 2 (mode discovery and credit assignment) and operation 4 (delta validation).

Meta-Harness Lee et al. [2026] is the cleanest representative. Its opening claim — that "the performance of large language model systems depends not only on model weights, but also on their harness: the code that determines what information to store, retrieve, and present to the model" — is §1's scaffold-versus-weights decomposition restated for the harness substrate specifically. Meta-Harness then identifies that existing text optimizers "compress feedback too aggressively," losing the fine-grained signal needed for credit assignment. Its solution is an outer-loop proposer with access to source code, scores, and execution traces of all prior candidates through a filesystem, so that the edit history itself becomes the credit-assignment signal. In Section III terms, the filesystem of prior candidates is a concrete form of the accumulated history H_t that conditions the proposal distribution Q_t, and the outer loop is a Loop-1 closure over the harness substrate (Section V, S5) rather than the skill substrate (S2).

Agentic Harness Engineering [AHE; Lin et al. [2026]] attacks the same problem from the observability side.

The core diagnosis is that harness engineering "faces a heterogeneous action space across editable components, voluminous trajectories that bury actionable signal, and edits whose effect is hard to attribute." These three failure modes map precisely onto Section III's mechanism stack: a heterogeneous action space is the coordinate-identification problem (E.1 op 2); voluminous trajectories that bury signal is the residual-capture problem (op 1); and effects that are hard to attribute is the credit-assignment half of op 2 (E.6). AHE's response — three matched observability pillars giving every editable harness component a file-level representation — is an engineering instantiation of the coordinate decomposition that Section III assumes. The implicit claim is that artifact-layer descent requires not just the optimization loop but also the instrumentation that makes scaffold coordinates individually observable and attributable.

MOSS Cai et al. [2026] extends the harness argument by observing that all prior self-evolving agent systems "confine evolution to text-mutable artifacts — skill files, prompt configurations, memory schemas, workflow graphs — and leave the agent harness untouched." Since routing, hook ordering, state invariants, and dispatch live in code rather than in any text artifact, an entire class of structural failure is, in MOSS's phrase, "physically unreachable from the text layer." MOSS therefore extends Loop-1 evolution to source-level code rewriting. In Section V terms, MOSS evolves S2 (Skills) and S5 (Orchestration) simultaneously, rather than treating the orchestration substrate as a fixed wrapper around an evolvable skill layer. This matters for Section IX: a governed cross-tenant scaffold optimizer that covers only text-mutable artifacts leaves the code substrate ungoverned, and MOSS's diagnosis suggests the gap is not merely inconvenient but structurally incomplete.

CANTANTE Zehle [2026] attacks credit assignment directly in the multi-agent setting. Its observation is that "scores are available only at the system level, whereas the parameters governing agent behavior are local," making optimization of multi-agent scaffolds "fundamentally a credit-assignment problem." CANTANTE's solution is contrastive: it decomposes system-level rewards into per-agent update signals by contrasting rollouts of multiple jointly acting configurations. In Section III vocabulary, CANTANTE provides a concrete mechanism for the credit-assignment step (E.1 op 2; E.6) in the case where the scaffold is distributed across agents rather than concentrated in a single artifact. The gap it closes is the same one Section III identifies in Appendix E.6 — that attribution (output error assigned to an upstream scaffold coordinate) must succeed with non-zero probability per failure — but instantiated for the multi-agent topology substrate (S5).

Harness optimizer evaluation Ong et al. [2026] provides independent empirical evidence for a failure mode this paper flags in Section X: that evaluating scaffold optimizers "solely by observing target agents' performance gains" misses intermediate erroneous edits that hinder performance. Their finding — that it is "unclear whether harness optimization is driven by optimizers' informed update actions or simply trial-and-error" — is a direct empirical instantiation of the distinction between Stage 2 (random artifact search) and Stage 3 (artifact-layer descent) from Section II's table. The credit-assignment problem is not merely theoretical: production harness optimizers that cannot distinguish informed updates from lucky trial-and-error are operating in Stage 2 even when they report benchmark improvements.

HARNESSEVO Nguyen et al. [2026] turns the same diagnosis into a measurement. It splits the harness into four separately evolvable slots (role, task strategy, tool and format rules, reflection and control) and attributes value by leave-one-in and leave-one-out evolution. On ALFWorld with a 7B target model nearly all of the gain lives in the reflection/control slot (leave-one-in +0.119; the other slots individually null), and dividing a fixed 64-rollout budget evenly across the four slots leaves each below the optimizer's effective search floor, so every slot freezes at its seed; concentrating the budget on the high-value slot recovers the gain (0.642 to 0.761). On WebShop the effect is absent. In Section III vocabulary this is direct evidence that credit assignment over scaffold coordinates must precede budget allocation, and that the coordinate decomposition of Section V is not merely descriptive: which coordinate carries value is task-dependent and must be measured, not assumed.

S5.7 Multi-objective and lifecycle constraints

Two papers expose constraints on the Section III picture that single-loss descent does not capture, both relevant to the Section IX missing-architecture discussion.

MOCHA Tanjim et al. [2026] observes that agent skills are "multi-field artifacts subject to hard platform

constraints: description fields are truncated for routing, instruction bodies are compacted via progressive disclosure, and co-resident skills compete for limited context windows." Skill optimization is therefore not a single-loss descent problem but an inherently multi-objective one: a skill must simultaneously maximize task performance and satisfy platform limits, and optimizing for task performance alone produces skills that pass benchmarks but violate deployment constraints. MOCHA introduces *Chebyshev annealing* to navigate the constraint frontier. In Section III terms, the complexity penalty $\Omega(z)$ in the gate condition is precisely the mechanism MOCHA is operationalizing, but Section III's scalar Ω is a simplification: in production, the constraint surface is a Pareto frontier, not a single penalty term. MOCHA's empirical contribution is to show that ignoring this frontier — treating skill optimization as unconstrained single-loss descent — produces skills that degrade on real deployment infrastructure even when benchmark scores improve.

Ratchet Zhang et al. [2026] makes an argument that is, in some respects, orthogonal to the gradient-analog framing and for that reason worth engaging directly. Ratchet's opening finding — that LLM-authored skills deliver +0.0 pp over no-skill baselines while human-curated ones deliver +16.2 pp — suggests that the bottleneck in self-evolving skill libraries is not the optimization algorithm (the mechanism for proposing and accepting edits) but *lifecycle management*: whether skills are written, retrieved, curated, and retired correctly. Ratchet introduces four hygiene mechanisms — outcome-driven retirement, a bounded active-cap, meta-skill authoring, and pattern canonicalization — and shows that these lift the +0.0 pp floor substantially before any gradient-analog optimizer is applied.

This finding is not a refutation of the Section III picture but a clarification of the precondition for it. Artifact-layer descent, as formalized in Section III, presupposes that the scaffold is maintained well enough for the gate G_t to have a meaningful signal: if skill retrieval is so noisy that the wrong skill is injected into every rollout, the proposal process Q_t receives corrupted evidence and the descent guarantee degrades to random search. Ratchet's hygiene mechanisms are, in Section III vocabulary, prerequisites for operation 1 (residual capture) and operation 2 (mode discovery) to function: outcome-driven retirement prevents stale artifacts from polluting the residual signal; the bounded active-cap prevents scaffold bloat from making retrieval unreliable (the Section X failure mode); pattern canonicalization collapses duplicate and near-duplicate skills so the proposal process conditions on a clean, deduplicated artifact set rather than competing variants of the same pattern. The practical implication is that the deployment path to Stage-3 artifact-layer descent runs through Stage-1 hygiene as a *prerequisite*, not as an alternative. A system with strong hygiene and no optimizer sits at Stage 1 with a clean scaffold; a system with a strong optimizer and poor hygiene sits at Stage 2 with a noisy loss signal. Stage 3 requires both.

SkillsVote Liu et al. [2026] carries the lifecycle argument to the scale of an open ecosystem. It profiles a million-scale open-source skill corpus for environment requirements, quality, and verifiability, synthesizes tasks for the verifiable subset, exposes skills through agentic library search before execution, and after execution decomposes each trajectory into skill-linked subtasks whose outcomes are attributed to skill guidance, agent exploration, environment, or result signals; only successful, reusable discoveries pass into evidence-gated updates. Two Section III points follow. First, attribution is performed at the admission gate rather than inside the optimizer, which is credit assignment (E.1 op 2) used as a filter on residual capture (op 1). Second, the redundancy and environment-sensitivity SkillsVote measures in open skill ecosystems are the same bloat and drift that Section X names, arising this time from many authors rather than one loop.

S5.8 Cross-task transfer and evidence radius

Several papers independently measure what Section III calls the *evidence radius* of a scaffold artifact — how far an edit validated in one context continues to improve performance when redeployed elsewhere — without using that vocabulary.

Training Language Agents to Learn from Experience [ICT; Shalev et al. [2026]] introduces the In-context Training framework specifically to evaluate *cross-task self-improvement*: whether experience distilled from one set of tasks transfers to unseen tasks. Their finding that "current reflection-based methods can only self-correct within a single task instance" is a measurement of evidence radius at its lower bound — a radius of exactly one task — and their ICT framework is a systematic attempt to expand that radius through supervised distillation of reflections into reusable lessons. In Section III terms, ICT is an attempt to

instrument Loop 2 empirically: to measure whether a locally-validated edit (one that passed the gate on one task's L_D) continues to improve performance when the promotion radius is expanded to cover a distribution of unseen tasks.

Combee Li et al. [2026] approaches the same measurement from the scaling direction. ACE Zhang et al. [2025] established that context-as-living-artifact (a system prompt that accumulates, refines, and organizes strategies through agentic interaction) could improve over static prompts on single-agent tasks. Combee extends this to high-parallelism settings, arguing that "existing methods primarily focus on single-agent or low-parallelism settings," which "fundamentally limits their ability to efficiently learn from a large set of collected agentic traces." The parallel scaling Combee addresses is a form of Loop-2 scope expansion: when many agents run simultaneously, the trace pool available for the next proposal step is drawn from a wider deployment distribution, and the promoted artifact is being tested against a larger averaging set of patch losses. This is not cross-organizational Loop-2 promotion, but it is a step toward it, and Combee's empirical results measure how much of the evidence-radius benefit can be captured by intra-organization parallelism before the cross-organizational barrier is reached.

CORAL Qu et al. [2026] addresses the Loop-2 problem from the multi-agent open-ended discovery angle. CORAL replaces rigid control with long-running agents that explore, reflect, and collaborate through shared persistent memory, asynchronous multi-agent execution, and heartbeat-based interventions. The shared persistent memory is the cross-agent promoted artifact: a finding validated in one agent's exploration that is written to the shared store becomes available to all agents, and the heartbeat mechanism provides a form of auto-gated Loop-2 promotion. CORAL does not implement the versioned lineage, RBAC, or rollback that Section IX requires, but it provides the only fully asynchronous multi-agent Loop-2 promotion mechanism in the current survey corpus, and its open-ended discovery framing makes the evidence-radius question concrete.

PACE Ling et al. [2026] closes this family with a production-oriented two-timescale design for small language model agents. The fast timescale adapts prompts, parsers, validators, and control logic from failure signals within a deployment; the slow timescale consolidates across episodes. The two-timescale structure maps cleanly onto the Loop-1 / Loop-2 distinction of Section III: Loop 1 fires at the fast timescale, updating a single deployment's scaffold; Loop 2 fires at the slow timescale, consolidating across the deployment's own episodes rather than across deployments. PACE's contribution is to show that the two-loop architecture is implementable under the resource constraints of small, frozen production models — not only under frontier-model assumptions — and that the slow/fast consolidation structure persists even when the optimizer model and the target model are the same frozen small model.

SkillClaw Ma et al. [2026] and **FederatedSkill** Yang et al. [2026a] are the two systems in the corpus whose Loop 2 crosses users rather than tasks or agents. SkillClaw treats cross-user, over-time interaction as the primary improvement signal: trajectories from many users are aggregated, an autonomous evolver identifies recurring behavioral patterns and translates them into skill-set updates, and the updated skills live in a shared repository synchronized across users; its evaluation protocol runs a day-night cycle in which simulated concurrent users generate trajectories by day and the system reflects, proposes bundles, validates, and deploys by night. FederatedSkill starts from the observation that isolated single-user task streams lack the diversity to build comprehensive skills, while raw trajectory sharing compromises privacy and a uniform global library ignores client heterogeneity. Its unit of communication is the semantic skill diff (a structured patch over a local library, shipped with the task reward that produced it); a server-side evolution agent aggregates patches into a per-client capability matrix (covered, absorbing, broken, gap) and emits a personalized update for each client, applying, adapting, or rejecting peer patches with an auditable decision log that records source patch, reward, and justification for every modified path. Across 20 task families it reports up to +44.4% success and -37.5% compute against isolated self-evolution. In Section III terms FederatedSkill is the first surveyed system in which Loop 2 is a federated aggregation of artifact updates rather than a promotion of a single artifact, and its decision log is the first lineage record at the aggregator. It remains inside one operator's federation (clients are backbone-plus-harness profiles, not organizations), its acceptance decision uses the source client's reward rather than a held-out evaluation at the receiving client, and it reports neither RBAC nor rollback; Section IX's properties (b), (c), and (d) are each partially or wholly open.

S5.9 Synthesis: what the convergence establishes

The four claims stated in Section VIII are supported as follows:

1. **The scaffold-versus-weights decomposition is independently confirmed.** GEPA, Meta-Harness, MOSS, RCL, and PACE all begin from the observation that model weights and the external artifact layer are distinct adaptation surfaces with different cost, reversibility, and scope properties. No surveyed paper disputes this decomposition; the variation is in which substrate each paper targets (skill document, harness code, system prompt, shared memory).

2. **Credit assignment is the central engineering bottleneck.** AHE, CANTANTE, Meta-Harness, Ong et al. [2026], HARNESSEVO, and SkillsVote all identify credit assignment — attributing a system-level failure to a specific scaffold coordinate — as the problem that distinguishes disciplined Loop-1 evolution from trial-and-error. This is the credit-assignment step of Appendix E.1 (operation 2; E.6), and it remains unsolved in the general multi-substrate case.

3. **Lifecycle hygiene is a precondition for descent, not an alternative to it.** Ratchet establishes that the gradient-analog mechanism of Section III requires a maintained scaffold as a precondition. A dirty artifact pool (stale, bloated, miscurated) prevents the gate G_t from receiving a meaningful signal. Hygiene and optimization are sequential dependencies, not competing paradigms.

4. **No surveyed system combines Loop-1 automation with cross-organizational Loop-2 governance.** SkillOpt has the strongest Loop-1 closure and the most systematic evidence-radius measurement, but it does not implement cross-organizational promotion, versioned lineage with RBAC, or rollback. CORAL implements asynchronous cross-agent promotion within one system but not across organizations. PACE implements two-timescale consolidation within one deployment. Meta-Harness implements Loop-1 closure over the harness substrate but no Loop-2 promotion mechanism. FederatedSkill federates skill patches across clients under one operator, with a decision log but no RBAC or rollback. The four-property gap named in Section IX is narrowed on property (b) by FederatedSkill and closed on none; it is, if anything, confirmed more strongly by the independent convergence of multiple research threads on the same single-organization ceiling.

S5.10 SkillOpt against Section IX's four properties

Mapped onto Section IX's four properties of governed cross-tenant scaffold optimization:

- (a) **Auto-gated Loop 1.** *Satisfied.* The held-out selection-split gate at strictly-greater margin instantiates G_t.
- (b) **Multi-tenant cross-org promotion.** *Absent.* SkillOpt is a single-organization research system; there is no governed cross-tenant aggregator.
- (c) **Versioned lineage with RBAC.** *Partial.* The exported `best_skill.md` is a versioned artifact, and the protected slow-update field separates fast and slow consolidation. There is no documented RBAC layer or audit-trail policy, and `edit_apply_report.json` is an operational log rather than an access-controlled lineage system.
- (d) **Rollback.** *Absent.* The training loop selects the best validation-gated skill and exports it; no rollback envelope on deployed artifacts is reported.

The cross-model, cross-harness, and cross-benchmark transfer experiments reported by SkillOpt are *empirical evidence-radius measurements*: they tell us that a skill optimized in one (model, benchmark, harness) cell continues to improve performance when redeployed in nearby cells without further optimization. In Section III vocabulary, this is a measured Loop-2 *radius* for the artifact, not a *governance mechanism* for promoting the artifact. Both readings are useful: SkillOpt is the strongest current Loop-1 proof and supplies the first systematic transfer evidence at the scaffold layer, while the Section IX cross-tenant governance triad ((b), (c), (d)) remains unfilled.

S5.11 Cost and bloat

SkillOpt reports 0.6M to 46.4M training tokens per absolute test-point gain (their Table 6), with the procedural benchmarks (SpreadsheetBench, OfficeQA, LiveMath) at the cheap end and search/visual benchmarks (SearchQA, DocVQA) an order of magnitude more expensive per point. The deployed artifact remains 379–1,995 tokens regardless. This is, to our knowledge, the first published cost-per-point figure for scaffold-layer descent, and it complements the Section X scaffold-bloat discussion (and the Ratchet hygiene-precondition finding walked in G.7): the textual learning rate η_t with cosine decay keeps the deployed skill compact even when the optimizer explores many more edits than are accepted (only 1–4 edits survive into the final skill across the six benchmarks). The mechanism that prevents the deployed artifact from bloating is exactly the complexity penalty $\Omega(z)$ of Section III, instantiated as a hard cap on accepted edits per step rather than a soft regularizer.

S6. Claims audit and the rubric-reward comparison

Every absolute claim in the arXiv short v3.2 (`arxiv/frontier_and_localhost/short/paper_frontier_and_localhost_sho` line numbers as of 2026-09-20) of the form *no surveyed system / only X / first / absent / empty cell*, with its verdict after the October-2026 literature sweep (three parallel sweeps: skills; harness, context, and LLMOps governance; memory governance, competing surveys, rubric rewards; 127 systems added, scored from their own method sections) and the recount in `scripts/recount.py`. Verdicts: **holds** (unchanged), **sharpened** (true, now stated more precisely), **revised** (wording must change), **dropped** (not carried into the IEEE version). Evidence names the system(s) that forced the verdict; arXiv ids are in `submission/references.bib`.

Corpus after the sweep: 256 unique systems (130 May + 126 October), 204 core, 52 contrast. No October addition scores 5 on any substrate. Composite two-loop audit: 14 research systems + 3 industry exemplars (Sierra Agent OS 2.0, Cognition Devin, and now Letta Context Repositories, whose shared-memory push carries git lineage but no gate) pass; 7 October systems promote across contexts without versioning or lineage and enter only under relaxation (a) or not at all (EvolveNet, Evo2Team, XRepoSkill, ShareMem, RASO, Spotify, Future AGI (T5 only)).

Corrections to the May data applied here (`scripts/apply_corrections.py`): (1) the row "Governed Collaborative Memory / SSGM" conflated Cuadros et al. (2605.04264) with SSGM (Lam et al., 2603.11768); split, SSGM scored 2 (conceptual framework). (2) MAE and Memento-Skills were scored 5 on S5 in the CSV while the published table said no orchestration system reaches 5; both close one loop only, so they are re-scored 4 per the uniform rubric. S5 mean for the May corpus moves from 2.68 to 2.58.

S6.The §9 corner (a)–(d)

Line	Claim	Verdict	Evidence
70, 120, 235, 246, 272, 574, 578, 592, 698	No public system combines (a) auto-gated Loop 1, (b) cross-organization promotion, (c) versioned lineage with RBAC, (d) rollback	**holds**	Three sweeps, 126 systems: (b) is unclaimed by any evidenced system (EvolveNet's clients are benchmark partitions, "organizations" only in motivation; MemClaw is single-tenant in its evaluation; Buildrix is an open community hub with no gate); (c) appears as lineage without RBAC everywhere (EDDOps, ACM, MemClaw, Letta, Mem0 history) and as RBAC without versioned promotion in MCP control planes (agentic-community gateway, TrueFoundry); (d) is demonstrated by Governed Capability Evolution and EDDOps and claimed by EvolveNet (round rollback) and EvoUndo (recoverability as admission), none of which has (a)+(b).
110	"No surveyed system meets the FedAvg limit; FederatedSkill comes closest"	**sharpened**	EvolveNet (2608.04968) is now the nearest: broadcast, client-local optimization, server-side merge accepted iff fixed $\geq$ broken on a held-out slice, rollback to the previous shared harness. Still selection-and-merge over program edits, not averaging; cite both.
155	"(iv) ... FederatedSkill comes closest ... narrowed on one property, closed on none"	**revised**	Two closest approaches now: FederatedSkill (decision log, no gate at receiver) and EvolveNet (held-out merge gate and round rollback, no lineage or RBAC). Say "narrowed on (b) and (d) by different systems, closed on none."

Line	Claim	Verdict	Evidence
242	"no member of the §3.5 research families provides (d), and only FederatedSkill provides even a partial (b)"	**revised**	(d): EvolveNet (round rollback), SkillSmith and SkillAudit (edit-level rollback inside one agent), Growing Harness and WDA (within-optimization rollback), EvoUndo (recoverability as an admission test). (b): EvolveNet and RASO partial. New sentence: "rollback now appears inside single deployments and inside one federated round; no system rolls back a promotion that has crossed an organization boundary, because none makes such a promotion."

S6.Matrix cells

Line	Claim	Verdict	Evidence
227	"Three cells are populated; the rest are empty for structural reasons"	**revised**	Among composite-passing systems the three cells hold; but single-loop and governance-only systems now occupy [Human × Human] (MCP registries: Azure API Center, agentic-community MCP Gateway & Registry, TrueFoundry; Buildrix; Enoch), [Human × None] (VerificAgent, revocation guard, Davis et al. governance conversion, Continual Harness human phase), [Auto+Human × None] (SkillChain, Learning on the Job), [None × Auto] (Governed Capability Evolution), [None × Auto+Human] (EDDOps), [None × Human] (ACM). Restate as: three cells hold every system that passes the composite audit; the human-authored cells are occupied only by registries and curation workflows with no session-signal loop.
229, 549	"[Human × Auto] is empty by engineering logic . . . no published example exists"	**holds**	No system with human-authored candidates and automated promotion was found; the closest, MCP Gateway & Registry, adds an automated security scan to a human admission, which is [Human(+scan) × Human].

Line	Claim	Verdict	Evidence
229, 551	"[Human × Human] is empty in the surveyed corpus"	**revised**	Occupied by MCP registries and a community hub; see above. The paper's predicted occupant (solo or small-team workflows) is also now documented empirically by "Who Maintains Agent Skills?" (2609.05677): across five public skill repositories every substantive edit is human-merged, 62% AI co-authored, 1 deprecation in 254 edits.
229	"[Human × None] is structurally improbable"	**revised**	Occupied: VerificAgent (agent-drafted memories, human fact-check, frozen bank), the revocation guard of Shen et al., and human-in-the-loop harness phases. Restate: it is the human-curation cell, single-loop by construction.
231	"Nearly every auto-gated system fires at n = 1"	**revised**	Batched or statistical acceptance is now common: SAGE-gate (paired per-item test), Future AGI (paired bootstrap CI), AutoSaddler (mini-batch failure signals), MGM, ModularRSI (evidence across tasks), VaG (3-of-3 critics), EvolveNet (fixed ≥ broken on held-out), VACE, HackProbe, PROCTOR. Say "common since mid-2026" and cite SAGE-gate and VaG.
231	"no surveyed system has formalized agreement as a k-of-n acceptance rule"	**revised**	VaG requires all three heterogeneous critics to pass (3-of-3); SkillSpec is consensus-gated. Keep the pˆk-vs-1/k distinction, drop "no surveyed system".

Line	Claim	Verdict	Evidence
231	"Only AlphaEvolve demonstrates explicit multi-layer depth"	**sharpened**	Still the only system with cascade evaluator plus diversity archive; HarnessBank, JIT-Agent, MGM keep archives (lineage), VaG adds tiered subset selection. Say "AlphaEvolve remains the only system combining a cascade evaluator with a diversity archive."

S6."Only" and "first"

Line	Claim	Verdict	Evidence
145, 299, 564, 682	CORAL "the only asynchronous multi-agent Loop-2 mechanism in the corpus"	**revised**	Evo2Team (2609.34135) promotes routing and communication skills across 4–32-agent teams with target-side confirmation; ShareMem pools memory across users. Say "CORAL was the only one in the May corpus; Evo2Team and ShareMem add cross-team and cross-user promotion without versioning."
213	AlphaEvolve Strassen claim	**holds** (attributed)	Third-party; attribute to the AlphaEvolve report in the same sentence.
215	"CASCADE reports the largest within-benchmark ablation gain in the corpus"	**dropped**	Not re-checked across 127 new systems; comparing ablation gains across benchmarks is not meaningful. Removed from the IEEE text.
254, 317	ChronoMem and rollback repair "the first surveyed instances of the rollback envelope"	**revised**	Rollback of memory is now in VerMem (soft-delete/restore ops), Mi-Memory (design), Letta (git history); and shown unsafe in checkpoint-rollback studies (2608.29381, 2608.15939). Say "among the first" and cite the counter-evidence.

Line	Claim	Verdict	Evidence
316	Cuadros "only surveyed memory system with full provenance + versioning + correction"	**holds**	Memory governance additions (TOKI, PPMF, TMA-NM, MemClaw, He & Yu) each carry provenance or lineage without correction loops or versioning, and the auto-gated memory loops (Recuris, MemGuard, VerMem, Compiled Memory) carry no lineage.
610, 630	SkillOpt "cleanest published instance" / "first to instantiate all four simultaneously"	**sharpened**	SkillOpt-Lite (2607.03451) removes two of the four and matches it; VALVE, SkillSmith, SkillAudit instantiate held-out gates with rollback. Say "the first system we found that instantiates all four, and its own follow-up shows two of them are stabilizers rather than necessities."
686	FederatedSkill "first ... federated aggregation ... first lineage record at the aggregator"	**holds**	EvolveNet (Aug 2026) postdates it and has no lineage record.
713	SkillOpt "first published cost-per-point figure"	**holds**	No earlier figure found.

S6.Absences

Line	Claim	Verdict	Evidence
62	"production deployments sit at Stage 1 or 2; none reaches Stage 3"	**sharpened**	Spotify (2609.30297) runs automated prompt and planning optimization gated by human launch and A/B metrics; SkillChain (Alibaba) runs judge-monotone edits behind a human reflection gate with reported agreement. Both are Stage 3 on Loop 1 with a human Loop 2; say "production deployments reach Stage 3 on the local loop only behind a human promotion gate".
188	S1 score-5 none; "no instruction system closes both loops"	**holds**	S1 n=15, max 4; adoption studies cut the other way (Gloaguen: context files raise cost >20% without raising success; Galster: context files dominate configuration, skills rare).
189	"Anthropic's skills repository is one-pool with PR review but no automated eval gate"	**holds**	Confirmed by 2609.05677 (human-merged, no eval gate observed).

Line	Claim	Verdict	Evidence
190	"correctable, versioned memory is rare; production memory lacks failure-triggered correction and versioned lineage"	**revised**	Letta Context Repositories (git-backed memory, every edit committed, shared memory auto-pushed), Mem0 per-memory history and feedback endpoint, Zep bitemporal invalidation: lineage is now common in production memory; failure-triggered correction is not, and 2609.08258 shows revoked facts still outrank their replacements in five backends. Say "versioned lineage has arrived in production memory; enforced correction has not."
192	S5 score-5 none; "topology evolution is rare"	**sharpened**	After the correction S5 has no 5 in either corpus. Gated topology evolution now exists: TOCOMAS retains only candidates satisfying structural invariants; Evo2Team promotes across teams; OptiMAS, AgentGrad, HarnessForge update S5 under evaluation. Say "rare and, where it exists, gated by evaluation alone except TOCOMAS."
196	"no system matures all six at once"	**holds**	No October addition scores 5 on any substrate.
198a	"Failure-triggered memory is rare in production"	**holds**	One field report (SIx Harness, N=1, self-reported); Letta dreaming writes from conversation history, not failures.

Line	Claim	Verdict	Evidence
198b	"Cross-customer skill promotion is nearly absent"	**holds**	ShareMem, EvolveNet, XRepoSkill, RASO promote across users, clients, or repositories inside one operator; none across customers or organizations with evidence.
198c	"Memory versioning is mostly unsolved outside Cuadros et al. and ChronoMem"	**revised**	See 190: Letta, Mem0 history, Zep bitemporal facts, TOKI audit rows, MemClaw supersession chains.
198d	"Scaffold-level curriculum is unspecified in nearly every surveyed system"	**sharpened**	Task-CoEvolve (variance-weighted task sampling) and AutoSaddler (mini-batch failure signals) specify it; still the exception.
198e	"Held-out evals at Loop-2 promotion are absent"	**revised**	Receiving-side evaluation now exists: ShareMem (evaluation tasks belong to a different user), XRepoSkill (independent verification in $\geq$2 repositories), Evo2Team (target-side confirmation; 4 of 22 transfers still fail on new graphs), VACE (candidate evaluated with the receiving model fixed). EvolveNet explicitly has none; Federated Agent Optimization (2610.01195) proposes a receiving-client gate without implementing it.
206, 213	research full-auto: "enterprise governance is uniformly absent"	**holds**	No October [Auto $\times$ Auto] system reports RBAC or audited promotion.
208, 217	governance-first: "no patch-local inner loop"	**holds**	EDDOps, ACM, GCE, MCP control planes all have Loop 1 = None.

Line	Claim	Verdict	Evidence
219	"AGENTS.md in over 60,000 repositories"	**revised**	Newer counts: 2,853-repository study (Galster), 128,018-project adoption study (Robbes: 22–29% of coding-agent projects), 36,710 repositories mined for loop engineering (Lulla 2608.21884). Replace the single count with these studies and the cost finding (Gloaguen).
264	"no surveyed system treats pruning or expiration as a first-class gated operation (outside Ratchet, SkillOpt, ChronoMem)"	**revised**	Retirement is a gated operation in at least eleven skill systems (Library Drift: retire when n $\geq$ 100 and contribution $\leq$ -0.10 with a cap of 50; SkillSmith; SkillOps; ASSAY; HiSME; SkeMex; MUSE-Autoskill; SkillMAS; SafeEvolve; Double Ratchet; SkillGen) and expiry is a product control in Mem0, Zep, and LangGraph Platform. Say "pruning and expiry became first-class in 2026; what remains rare is pruning of a promoted artifact's descendants, which VaG shows is structurally irreversible."
266	failure surface "largely unmitigated in production"	**holds**	Revocation unenforced in five production backends (2609.08258); checkpoint rollback unsafe in three frameworks (2608.29381); no evidentiary record in 16 deployed frameworks (Hearsay 2609.32495).
375	multi-sample agreement "under-explored; SkillOpt-Lite nearest"	**revised**	See 231: SAGE-gate, VaG, SkillSpec.

Line	Claim	Verdict	Evidence
692	"No surveyed paper disputes the scaffold-vs-weights decomposition"	**holds**	SIA (2605.27276) updates harness and weights together but keeps them distinct levers.
694	credit assignment "remains unsolved in the general multi-substrate case"	**holds**	HARNESSEVO (slot attribution), AgentGrad (intervention-based), CANTANTE (contrastive) each solve one substrate's case.

S6.Section IV claims (checked against the rescored corpus)

Claim	Statement	Verdict	Evidence
IV-1	Every [Auto × Auto] system with a per-edit estimator gate runs on a C1 signal or a simulated C2 environment; [Auto × Auto] systems in C4 domains gate by proposal pruning	**holds**	May [Auto × Auto]: DGM, Voyager, AlphaEvolve, SkillOpt, SkillRL, SAGE, EvolveR, CoEvoSkills, AutoAgent, AutoManual (C1/simulated C2); NanoResearch (semantic merge, C4) and CORAL (heartbeat, C4) prune. October: EvolveNet, Evo2Team, XRepoSkill (C1 code/benchmark signals); ShareMem (C2 Mind2Web); RASO (target execution feedback, C1/C2); Letta (no gate).
IV-2	[Auto × Human] is populated only by C2/C3 systems	**holds**	SkillForge (C3 judge, human Loop 2), CASCADE (C3), Sierra (C2/C3), Devin (C2 merge outcome), Spotify (C2 A/B behind human launch). No C1 system occupies the cell.

Claim	Statement	Verdict	Evidence
IV-3	Rubric-as-reward builds C3 gates on the weight side; scaffold-side judge calibration under the reflexive condition is rare	**sharpened**	14 reward papers: 8 fix the rubric or judge during policy optimization, 5 co-evolve it (RLPR by construction, EvoRubrics, SCORE, RubricEM, Rubric-ARM at the reward-model stage); only 5 report any human-agreement number and none of the co-evolving ones does. Scaffold side: SkillForge (>90% agreement), SkillChain (κ 0.72–0.91), SkillCoach (trajectory-filter consistency 82→96% against human gold) now report calibration; none calibrates a judge that is itself being edited.

S6.Recount output (scripts/recount.py, merged corpus, all rows)

```
rows: 269  unique systems: 256  core: 204  contrast: 52
substrate            n   mean  score-5 systems
INTEGRATED           52   3.81  AutoAgent, NanoResearch, SkillRL
S1_Instructions      15   3.07  (none)
S2_Skills            50   3.34  COSPLAY, SAGE
S3_Memory            50   3.06  Governed Collaborative Memory (Cuadros et al.)
S4_Tools             20   3.65  Harvey, Hippocratic AI, MCP
S5_Orchestration     37   3.08  (none)
S6_Governance        45   3.02  AGENTS.md / AAIF, Braintrust, LangSmith Hub, Vellum
May corpus alone (after corrections): 130 / 99 / 31; S5 mean 2.58.
```

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